Cheng Chen

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Education

Institute of Automation, Chinese Academy of Sciences — Beijing, China
Incoming Ph.D. Student — Sept. 2027 (expected)
Supervisor: **Prof. Kaiqi Huang**

Tongji University — Shanghai, China
B.Eng. in Communication Engineering — Sept. 2023 - Jun. 2027 (expected)
Major rank: **2/85**

Research Interests

Multimodal learning and audio-visual understanding; large language models and multimodal large language models; embodied AI and reinforcement learning; efficient AI, knowledge distillation, model compression, and lightweight deployment.

Internships

Research Intern
Hong Kong University of Science and Technology (Guangzhou) — Sept. 2026 - Present
Zhejiang University — Jun. 2026 - Sept. 2026
Tongji University — Oct. 2025 - Jun. 2026

Academic Service

Reviewer: ICLR, PRCV, ICME, IJCNN, AAAI.

Publications

* Equal contribution.

- **Cheng Chen***, Yi Xu*, Wenzhuo Lei.
  If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization. *ACM MM 2026*, accepted.
- Yi Xu*, **Cheng Chen***, Mufan Cao.
  OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment. *IJCAI 2026*, accepted.
- Wenzhuo Lei, Mufan Cao, Chaorong Ye, Yi Xu, **Cheng Chen**.
  Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions. *IJCNN 2026*, accepted. **Corresponding author.**
- **Generated Image Evaluation**. *AAAI 2027*, under review, Phase 2.
  Assessing whether generated images satisfy textual instructions.
- **Reliability in Model Interpretability**. *ICLR 2027*, submitted.
  Evaluating the evidence behind explanations of neural-network behavior.
- **Understanding and Generation in Multimodal Models**. *ICLR 2027*, submitted.
  Studying how visual understanding relates to image-generation behavior.

Selected Research Experience

OV-OrthKD: Open-Vocabulary Audio-Visual Event Localization Co-first author

- Separated strong visual supervision from weaker audio supervision using orthogonal feature paths, with text prototypes for open-vocabulary transfer.
- Reached **0.816 segment AP** on OV-AVEBench. F1@0.5 improved by **2.7 points** overall and **3.4 points** on unseen categories over the official fine-tuning baseline.

OrthKD: Lightweight Diabetic Retinopathy Screening Co-first author

- Transferred complementary knowledge from heterogeneous teachers while restricting weaker-teacher supervision to features and reducing redundancy through orthogonality constraints.
- Evaluated on **132,049** retinal images. A **5.4M**-parameter student achieved **0.885 QWK** on EyePACS and improved zero-shot Messidor-2 QWK from **0.507 to 0.728** (quadratic weighted kappa).

DeepYOLO: Symbolic Visual Reasoning Corresponding author

Proposed the initial research idea and collaborated with a junior student, who led implementation and experiments, on converting images into scene descriptions for text-only language models.

Selected Software and Engineering Projects

AutoSearch AI GitHub

Citation-aware AI search with an arXiv paper radar and PDF reader. Combines a FastAPI backend with a Next.js frontend, streaming answers, and API/pipeline tests.

OfferPilot GitHub

Technical-interview practice prototype with multi-turn follow-up questions, per-question scoring, and structured reports. Uses TypeScript and Next.js, with configurable model providers and local session history.

FPGA Algorithm Optimization GitHub

Vitis HLS competition work targeting Zynq-7000 hardware. Includes C++ algorithm sources and design reports covering compression, cryptographic, and matrix-factorization kernels.

Research Skills GitHub

Curated workflow toolkit for CS/EE research, spanning experiment planning, paper implementation, evaluation, and hardware development. Includes an installable catalog and automated repository validation.

Selected Honors and Awards

- National First Prize, China Robot and Artificial Intelligence Competition (2026); Team Leader.
- National Second Prize, Global Campus AI Algorithm Elite Competition (2025); Team Leader.
- National Third Prize, Embedded Chip and System Design Contest, FPGA Track (2025); Team Leader.
- Provincial Second Prize, National College Student Mathematics Competition (2024–2025).
- Tongji University Outstanding Student Scholarship, Second Class (2023–2024 and 2024–2025).

Technical Skills

Programming: Python, C/C++, Verilog. **Machine learning:** PyTorch.
Tools: Git, Linux, LaTeX, OpenAI Codex, Claude Code.
Foundations: Machine learning, algorithms, linear algebra, probability, optimization, and signal processing.